

Heat conductance in infinite space

Solve $\partial_t u = D \partial_x^2 u$ (1)

for $x \in \mathbb{R}$, and $u(x, 0) = f(x)$

$$u(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i x \cdot k} \hat{u}(k, t)$$

$$\partial_t u(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i x \cdot k} \partial_t \hat{u}(k, t)$$

$$\partial_x^2 u(x, t) = \int \frac{dk}{\sqrt{2\pi}} (i k)^2 e^{i x \cdot k} \hat{u}(k, t)$$

$\Rightarrow$ (1) reads

$$\int \frac{dk}{\sqrt{2\pi}} e^{i x \cdot k} (\partial_t \hat{u}(k, t) + D k^2 \hat{u}(k, t)) = 0$$

i.e., the inv. F.T. of $\partial_t \hat{u}(k, t) + D k^2 \hat{u}(k, t)$ is identically 0 $\Rightarrow$

$$\partial_t \hat{u}(k, t) + D k^2 \hat{u}(k, t) = 0$$

This we can solve $\Rightarrow$

$$\hat{u}(k, t) = \hat{u}_0(k) \cdot e^{-D \cdot k^2 \cdot t}$$

solution:

$$v(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \hat{U}_0(k) e^{-D \cdot k^2 \cdot t}$$

$\hat{U}_0(k)$ is determined
by initial condition

$$f(x) = v(x, 0) = \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \hat{U}_0(k)$$

$$\Rightarrow \hat{U}_0(k) = \hat{f}(k) = \int \frac{dx'}{\sqrt{2\pi}} e^{-i \cdot k \cdot x'} f(x')$$

$$v(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \hat{f}(k) e^{-D \cdot k^2 \cdot t}$$

or, alternatively:

$$\begin{aligned} v(x, t) &= \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \int \frac{dx'}{\sqrt{2\pi}} e^{-i \cdot k \cdot x'} f(x') e^{-D \cdot k^2 \cdot t} \\ &= \int dx' \underbrace{\left(\int_{-\infty}^{\infty} \frac{dk}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi}} e^{i \cdot k \cdot (x - x')} e^{-D \cdot k^2 \cdot t} \right)}_{G_t(x - x')} f(x') \end{aligned}$$

G is referred to as Green's function

$$G_t = \mathcal{F}^{-1} \left[\frac{1}{\sqrt{2\pi}} e^{-D \cdot k^2 \cdot t} \right] = \frac{1}{\sqrt{2\pi}} \underbrace{\mathcal{F}^{-1} \left[e^{-\frac{1}{2} k^2 (2 \cdot D \cdot t)} \right]}_{\frac{1}{\sqrt{2Dt}} e^{-\frac{1}{2} \frac{x^2}{2 \cdot D \cdot t}}}$$

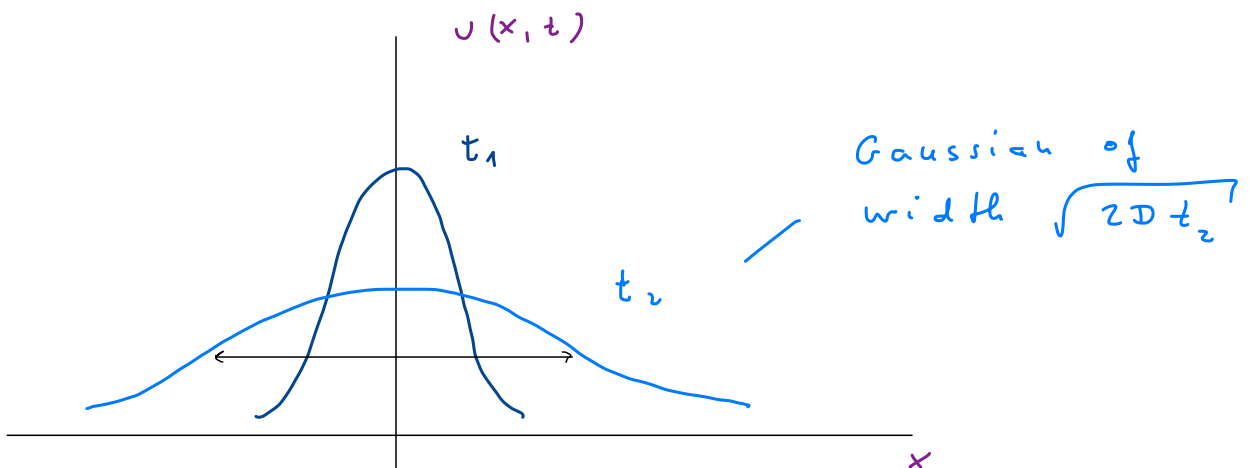
$$G_t(x) = \frac{1}{\sqrt{4\pi \cdot D \cdot t}} e^{-\frac{x^2}{4 \cdot D \cdot t}}$$

$$u(x, t) = \int dx' G_t(x - x') f(x')$$

consider initial condition, $f(x) = \delta(x)$

then:

$$u(x, t) = G_t(x)$$



$$\text{height} \sim \frac{1}{\sqrt{t}}$$

Complex functions

Lectures 12-13

Complex functions

basic notions

complex function

$$f: \mathbb{C} \rightarrow \mathbb{C}$$

$$f(z) = w$$

$$x + iy$$

$$u(x, y) + i v(x, y)$$

one complex function $\Leftrightarrow$ two $\mathbb{R}^2 \rightarrow \mathbb{R}$ functions

- limit ℓ is the limit of f at z_0
if $\forall \epsilon \in \mathbb{R} \exists \delta$ such that for $\forall z \neq z_0$
 $|f(z) - \ell| < \delta$ if $|z - z_0| < \epsilon$

- continuity: $f(z)$ is continuous at z_0 , if
its limit exists and is $f(z_0)$

- domain: typically an open subset D of $\mathbb{C}$



- f is continuous on domain D if it is
continuous at any point $z \in D$

Differentiation: f is differentiable at z

$$\text{if } \exists \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = f'(z)$$

example: $f(z) = z^2$

$$f(z + \Delta z) - f(z) = (z + \Delta z)^2 - z^2 = 2z \cdot \Delta z + \Delta z^2$$

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z$$

Def: • $f(z)$ is analytic at z if it is differentiable in a neighborhood of it

• $f(z)$ is analytic in a domain D if it is differentiable at any $z \in D$

Differentiability has serious implications...

Thm: Cauchy - Riemann theorem

f is differentiable at $z_0 \Leftrightarrow$ the partial differentials $\partial_x u, \partial_y u, \partial_x v, \partial_y v$ exist, and

$$(1) \quad \partial_x u = \partial_y v, \quad (2) \quad \partial_y u = -\partial_x v$$

$$(\Rightarrow) \quad \exists \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) \text{ holds when we approach}$$

z_0 from ANY direction

choice a: $z = z_0 + \Delta x$ then

$$f'(z_0) = \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) + i v(x_0 + \Delta x, y_0) - u(x_0, y_0) - i v(x_0, y_0)}{\Delta x} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x}$$

$$f'(z_0) = \partial_x u|_{x_0, y_0} + i \partial_y v|_{x_0, y_0}$$

choice b : $z \equiv z_0 + i \Delta y$, then

$$f'(z_0) = \lim_{\Delta y \rightarrow 0} \left[\frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{i \cdot \Delta y} + \frac{v(x_0, y_0 + \Delta y) - v(x_0, y_0)}{\Delta y} \right]$$

$$= \partial_y v - i \partial_y u$$

equating the two expressions for $f'(z_0)$ yields

$$\partial_x u = \partial_y v \quad ; \quad \partial_x v = -\partial_y u$$

($\Leftarrow$)

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} =$$

$$z = z_0 + \Delta x + i \Delta y$$

$$= \lim_{\Delta x + i \Delta y} \left[\frac{u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0)}{\Delta x + i \Delta y} + i \frac{v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0)}{\Delta x + i \Delta y} \right]$$

$$= \frac{\partial_x u \cdot \Delta x + \partial_y u \cdot \Delta y + \dots}{\Delta x + i \Delta y} + i \frac{\partial_x v \cdot \Delta x + \partial_y v \cdot \Delta y + \dots}{\Delta x + i \Delta y}$$

$$= \frac{\partial_x u}{\Delta x + i \Delta y} (\Delta x + i \Delta y) + i \frac{\partial_x v}{\Delta x + i \Delta y} (\Delta x + i \Delta y)$$

$$\uparrow$$

$$= \lim_{\Delta x + i \Delta y} \left[\partial_x u|_0 \cdot (\Delta x + i \Delta y) + i \partial_x v|_0 \cdot (\Delta x + i \Delta y) \right]$$

Cauchy
- Riemann

$$= \partial_x u + i \partial_x v = f'(z_0) \quad \checkmark$$

□

check a few functions

$$\bullet \quad f(z) = z^2 = \underbrace{x^2 - y^2}_u + i \underbrace{2xy}_v$$

$$\partial_x u = 2x = \partial_y v \quad \checkmark$$

$$\partial_y u = -2y = -\partial_x v \quad \checkmark$$

$$\bullet \quad f(z) = \bar{z} = x - iy$$

$$\partial_x u = 1 \neq \partial_y v = -1 \quad !$$

$z \rightarrow \bar{z}$ is NOT analytic

$$\bullet \quad e^z = e^x (\cos y + i \sin y)$$

$$u = e^x \cos y \quad v = e^x \sin y$$

$$\partial_x u = e^x \cos y \quad \partial_y v = e^x \cos y \quad \checkmark$$

$$\partial_y u = -e^x \sin y = -\partial_x v \quad \checkmark$$

e^z is analytic!

Corollary: $f(z)$ and $f'(z)$ differentiable in D

$\Rightarrow$ u and v satisfy the 2D Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

proof: f differentiable $\Rightarrow \partial_x u = \partial_y v$

$\Rightarrow$ differentiating:

$$\partial_x^2 u = \partial_x \partial_y v = \partial_y \partial_x v = -\partial_y \partial_y u$$

$$\Rightarrow \partial_x^2 u + \partial_y^2 u = 0 \qquad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

similar derivation for v

example $u = x^2 - y^2$

$$\frac{\partial^2 u}{\partial x^2} = 2 = -\frac{\partial^2 u}{\partial y^2} \quad \checkmark$$

Rules of differentiation:

carry over ...

- $(\alpha f(z) + \beta g(z))' = \alpha \cdot f'(z) + \beta \cdot g'(z)$
- $(f \cdot g)' = f' \cdot g + f \cdot g'$
- $\left(\frac{1}{f}\right)' = -\frac{f'}{f^2}$
- $f(z) = w \qquad f^{-1}(w) = z \qquad \Rightarrow$

$$(f^{-1})'(w) = \frac{1}{f'(z)} \qquad \left(\frac{dz}{dw} = \frac{1}{\frac{dw}{dz}} \right)$$

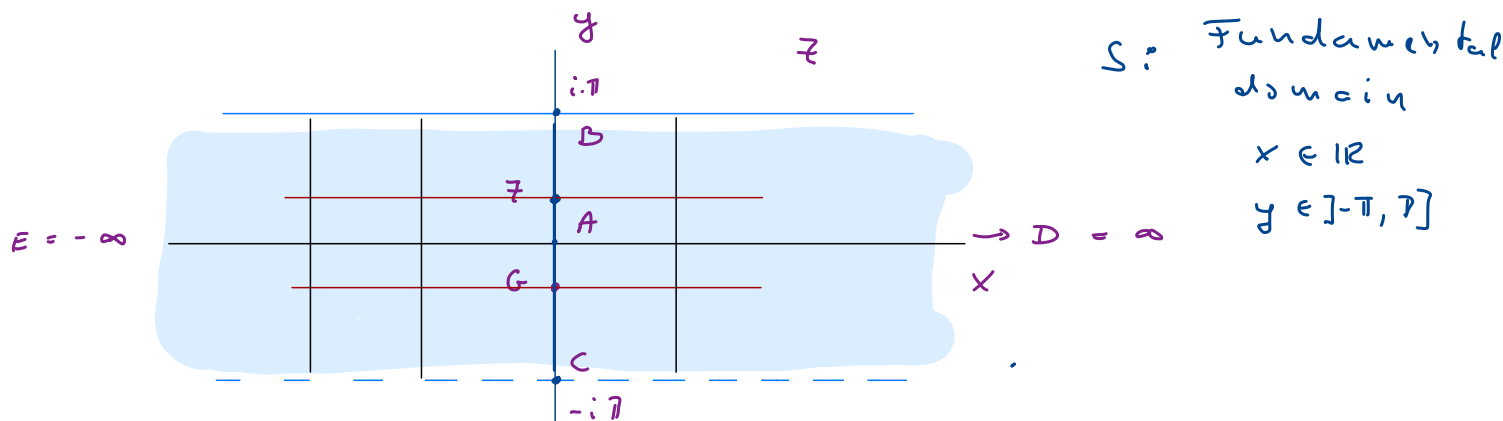
inverse function theorem

Complex Ln function

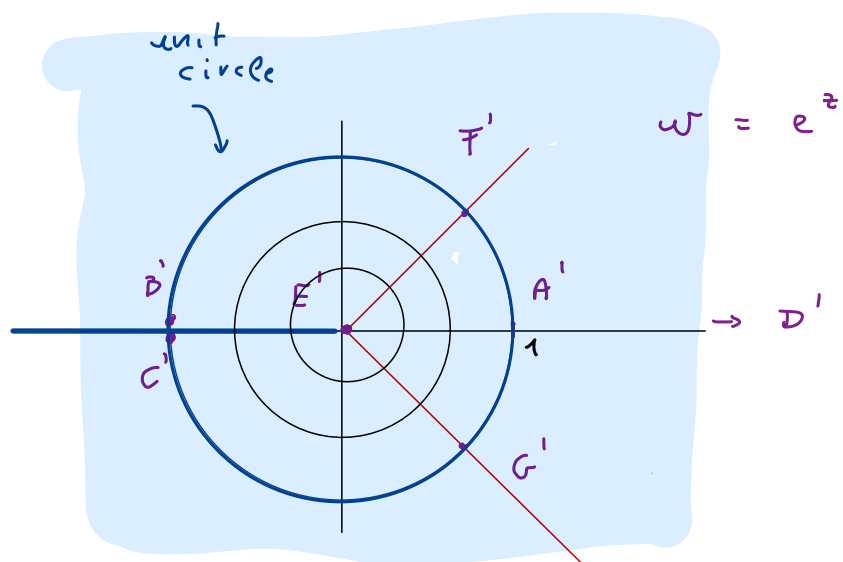
consider $f(z) = e^z = e^x (\cos(y) + i \sin(y))$

clearly $e^z = e^{z + 2\pi i \cdot n} \quad n \in \mathbb{Z}$

→ takes all its possible values on a strip



Construct image of this strip S



$$\begin{aligned} A' &: e^0 = 1 \\ B' &: e^{i\pi} = -1 \\ C' &: e^{-i\pi} = -1 \\ D' &: e^\infty = \infty \\ E' &: e^{-\infty} = 0 \end{aligned}$$

$$f: S \rightarrow \mathbb{C} !$$

remark: horizontal lines → rays
vertical lines → circles

e^z is angle-preserving!
= holomorphic

Statement: complex analytic functions
are holomorphic! ~ locally shape-preserving

$f = e^z$ is invertible on S !

$$e^z = e^x (\cos(y) + i \sin(y)) \equiv w$$

$$x = \ln e^x = \ln |w|$$

$$y = \operatorname{Arg}(w) \leftarrow \in]-\pi, \pi]$$

Def: $\operatorname{Ln}(w) = \ln |w| + i \operatorname{Arg}(w)$

$$\operatorname{Ln}: \mathbb{C} \rightarrow S$$

Remark: Ln is analytic!

(inverse of a differentiable function is also differentiable!)

Q: What is $\operatorname{Ln}'(w)$?

$$f(z) = e^z = w \rightarrow f^{-1}(w) = \operatorname{Ln}(w) = z \in S$$

$$\operatorname{Ln}'(w) = (f^{-1})'(w) = \frac{1}{f'(z)} = \frac{1}{e^z} = \frac{1}{w}!$$

Complex cos, sin, sh, ch functions

simple extensions of real functions :

$$\operatorname{ch}(z) \equiv \frac{1}{2} (e^z + e^{-z})$$

$$\operatorname{sh}(z) \equiv \frac{1}{2} (e^z - e^{-z})$$

$$\cos(z) \equiv \frac{1}{2} (e^{iz} + e^{-iz})$$

$$\sin(z) \equiv \frac{1}{2i} (e^{iz} - e^{-iz})$$

They can be expressed in terms of x, y :

Consider, e.g., $\operatorname{ch}(z)$

$$\operatorname{ch}(z) = \frac{1}{2} (e^z + e^{-z}) =$$

$$= \frac{1}{2} (e^x (\cos(y) + i \sin(y)) + e^{-x} (\cos(y) - i \sin(y)))$$

$$= \operatorname{ch}(x) \cdot \cos(y) + i \sin(y) \operatorname{sh}(x)$$

• addition rules survive :

$$\text{e.g.: } \operatorname{ch}(z_1 + z_2) = \frac{1}{2} (e^{z_1 + z_2} + e^{-(z_1 + z_2)}) =$$

$$= \underbrace{\frac{e^{z_1} + e^{-z_1}}{2}}_{\operatorname{ch}(z_1)} \underbrace{\frac{e^{z_2} + e^{-z_2}}{2}}_{\operatorname{ch}(z_2)} + \underbrace{\frac{e^{z_1} - e^{-z_1}}{2}}_{\operatorname{sh}(z_1)} \underbrace{\frac{e^{z_2} - e^{-z_2}}{2}}_{\operatorname{sh}(z_2)}$$

$$= \operatorname{ch}(z_1) \operatorname{ch}(z_2) + \operatorname{sh}(z_1) \operatorname{sh}(z_2)$$

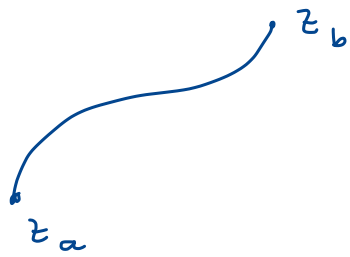
$$\bullet \quad \operatorname{sh} \leftrightarrow \sin : \quad \operatorname{sh}(iz) = i \sin(z)$$

$$\operatorname{ch}(iz) = \cos(z)$$

$$\bullet \quad \tanh(z) = \frac{\sin(z)}{\cos(z)} \quad , \quad \operatorname{th}(z) = \frac{\operatorname{sh}(z)}{\operatorname{ch}(z)} \quad \checkmark$$

Complex integration

Consider a curve $C : z(t) \quad t \in [a, b]$



$$z(t) = x(t) + i y(t)$$

↑
continuous

piecewise differentiable

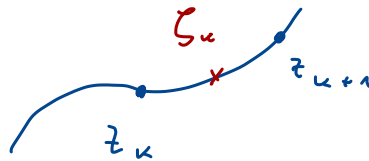
divide $[a, b]$ into n segments

$$t_0 = a < t_1 < t_2 \dots < t_n = b$$

correspondingly

$$z_k \equiv z(t_k)$$

define internal point



define sum:

$$S_n \equiv \sum_{k=0}^{n-1} f(\zeta_k) \Delta z_k$$

$\downarrow$
 $z_{k+1} - z_k$

$$\Delta t \equiv \max_k \{t_{k+1} - t_k\}$$

Def:

$$\int_C dz f(z) \equiv \lim_{\substack{\Delta t \rightarrow 0 \\ n \rightarrow \infty}} S_n$$

Q1: is it defined?

Q2: dependence on C ? $\leftrightarrow$ $\exists$ primitive function?

Q1: $z(t)$ continuous $\Rightarrow \Delta t \rightarrow 0$ implies $\Delta z \rightarrow 0$
 $\sim \Delta x, \Delta y \rightarrow 0$

$$\begin{aligned} S_n &= \sum_k (u_k + i v_k) (\Delta x_k + i \Delta y_k) = \\ &= \sum_k \Delta x_k \cdot u_k - \sum_k \Delta y_k \cdot v_k \\ &\quad + i \sum_k \Delta x_k \cdot v_k + i \sum_k \Delta y_k \cdot u_k \end{aligned}$$

taking $\Delta t \rightarrow 0 \sim \Delta x$ and $\Delta y \rightarrow 0$ these sums converge to line integrals for $u(x,y)$ and $v(x,y)$ piecewise continuous:

e.g. $\lim_{\Delta x \rightarrow 0} \sum_k \Delta x_k \cdot u_k = \int_C dx \, u(x,y) =$
 $= \int_a^b dt \, \dot{x}(t) u(x(t), y(t))$

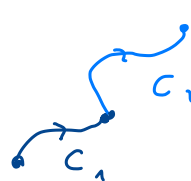
Thus

$$\int_C dz \, f(z) = \int_C dx \, u - \int_C dy \, v + i \left(\int_C dx \cdot v + \int_C dy \cdot u \right)$$

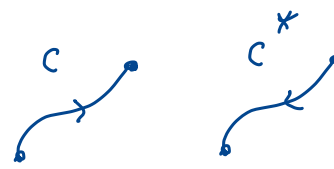
Properties:

- Linear:

$$\int_C dz (\alpha f(z) + \beta g(z)) = \alpha \int_C dz f(z) + \beta \int_C dz g(z)$$

- 

$$\int_{C_1} dz f(z) + \int_{C_2} dz f(z) = \int_{C_1 + C_2} dz f(z)$$

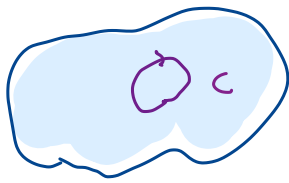
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$$\int_C dz f(z) = - \int_{C^*} dz f(z)$$

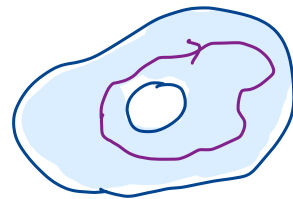
A practical formula:

$$\int_C dz f(z) = \int_a^b dt \underbrace{\dot{z}(t) f(z(t))}_{\mathbb{R} \rightarrow \mathbb{C} \text{ function}}$$

Def: simply connected domain: a domain in which any closed curve can be shrunk to a point.



simply
connected



NOT
simply
connected

Theorem 1: $f(z)$ differentiable on a simply connected domain and C non-intersecting piecewise continuous $\Rightarrow$

- $\int_C f(z) dz$ is independent of C

- $\exists F(z)$ such that

$$\int_{z_a}^{z_b} f(z) dz = F(z_b) - F(z_a)$$

- and $F'(z) = f(z)$

This is related to Cauchy's integral thm.:

Thm. 2 (Cauchy): C closed, single-valued piecewise differentiable curve on a simply connected domain, D , and $f(z)$ a differentiable (analytic) function

$$\Rightarrow \oint_C dz f(z) = 0$$

Some examples:

- e^z analytic on $\mathbb{C} \Rightarrow \oint_C dz e^z = 0 \quad \forall C$

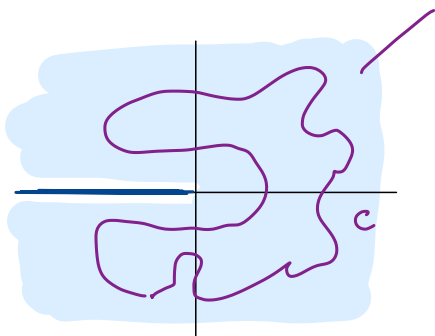
- $f(z) \equiv \bar{z}$ not analytic!

consider $z(t) = \rho e^{it} \quad t \in [0, 2\pi]$

$$\begin{aligned} \oint dz \bar{z} &= \int_0^{2\pi} dt \frac{dz}{dt} \cdot \bar{z} = \rho^2 \int_0^{2\pi} dt \cdot i e^{it} e^{-it} \\ &= 2\pi \cdot i \rho^2 \neq 0 \end{aligned}$$

- $\text{Ln}(z)$

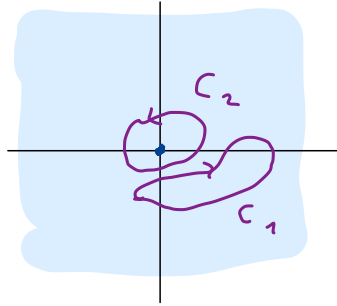
D simply connected



$$\oint_C \text{Ln}(z) dz = 0 !$$

• $f(z) = \frac{1}{z}$ analytic on $\mathbb{C} \setminus \{0\}$

not simply connected



consider $z(t) = g e^{it} \leftrightarrow C_2$

$$\oint_{C_2} dz f(z) = \int_0^{2\pi} dt \cdot g i e^{it} \cdot \frac{1}{g e^{it}} = 2\pi \cdot i !$$

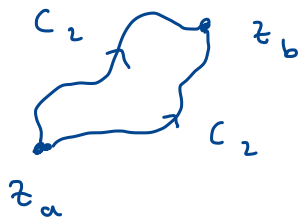
independent of shape of C_2 !

but!! $\oint_{C_1} f(z) dz = 0 !$

can be shrunk to $\emptyset$!

Thm. 1 follows from Thm. 2 :

D simply connected, C_1 and C_2 curves from $z_a \rightarrow z_b$ in D :

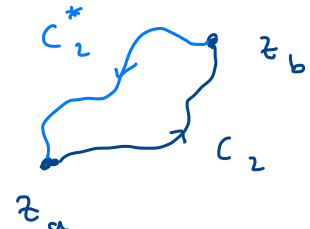


$$\int_{C_1} dz f(z) - \int_{C_2} dz f(z) =$$

$$- \int_{C_2^*} dz f(z)$$

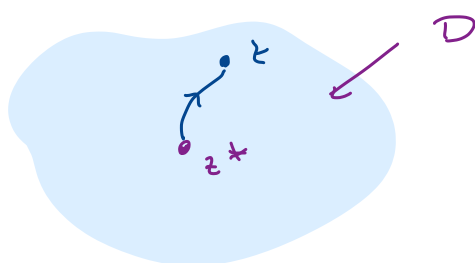
$$= \int_{C_1} dz f(z) + \int_{C_2^*} dz f(z) = \oint_{C_1 + C_2^*} f(z) dz \stackrel{\text{Cauchy}}{=} 0$$

$$\Rightarrow \int_{C_1} dz f(z) = \int_{C_2} dz f(z) \quad \checkmark$$



□

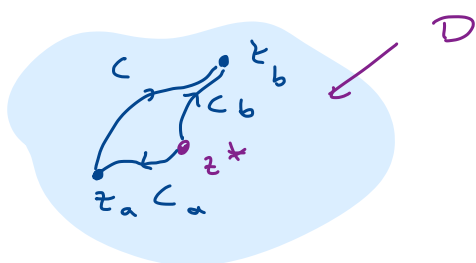
Now I construct $F(z)$!



$$z^* \in D \text{ (simply connected)}$$

$$\Rightarrow F(z) \equiv \int_{z^*}^z dz f(z)$$

↑
any curve



$$\int_{C_a} dz f(z) = \int_{C_a^* + C_b} dz f(z) =$$

$$= - \int_{C_a} dz f(z) + \int_{C_b} dz f(z) = F(z_b) - F(z_a)$$

Finally, we must show $F'(z) = f(z)$

$$\frac{1}{\Delta z} (F(z + \Delta z) - F(z)) = \frac{1}{\Delta z} \int_z^{z + \Delta z} dw f(w)$$

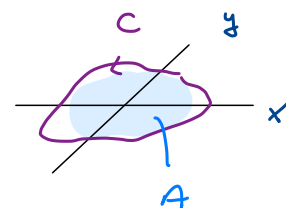
in the limit $\Delta z \rightarrow 0$ the integrand on the right converges to $f(z)$, and the integral to $\Delta z \cdot f(z)$ ✓

Proof of theorem 2

Consider, e.g., imaginary part:

$$\text{Im} \oint_C f(z) dz = \oint_C dx \underbrace{v}_{F_x} + \int_C dy \cdot \underbrace{u}_{F_y}$$

$$= \oint d\vec{\ell} \cdot \vec{F}(x)$$



Green (Stokes) theorem,

$$\oint_{C=\partial A} d\vec{\ell} \cdot \vec{F}(x) = \int_A dA (\partial_x F_y - \partial_y F_x) = 0 \quad \checkmark$$

$\partial_x u - \partial_y v = 0$ (if differentiable)